1 Code overview

`models.py`

Implements the five architectures required in Q 3: **Canonisation_Net**, **Symmetrization_Net**, **Sampled_Symmetrization_Net**, **Linear_eq_Net** (built from two equivariant layers), and **AugmentedInvariantNet**. Each class exposes `forward(X)` with $X \in \mathbb{R}^{N \times d}$.

`utils.py`
- Permutation helpers: `create_permutations_sampled`.
- *Unit tests* for invariance/equivariance (`test_canonization_net`, ...).
- Training routine `train_variance_net` that learns variance purely from permutation augmentations.

`main.py`

Orchestrates the workflow:

(a) runs each invariance test 50 times and prints the failure rate;

(b) trains the variance model for 200 epochs on a fixed set of size $N = 50,\ d = 5$;

(c) re-tests the trained model for invariance.

Dependencies are limited to `torch` and `numpy`; the script detects a GPU via `torch.cuda.is_available()` and moves tensors accordingly.

2 Permutation–invariance sanity check

test Permutation invariance

For every network we apply 50 random permutations $P \in S_N$ to a fixed set X and record a *failure* when the ℓ_∞–distance between the original and permuted outputs exceeds a model-specific tolerance:

(a,b,d) fail if $\|f(X) - f(PX)\|_\infty > 10^{-5}$;

(c) fail if $\|f(X) - f(PX)\|_\infty > 2 \times 10^{-1}$
(higher threshold to reflect the Monte-Carlo approximation of S_N);

(e) fail if $\|f(X) - f(PX)\|_\infty > 5 \times 10^{-2}$
(relaxed tolerance to probe the invariance learned via augmentation).

Finally, for each model we report the *failure rate*

$$\text{Failure rate} = \frac{\#\ \text{failed permutations}}{\#\ \text{permutations tested}},$$

i.e. the fraction of the 50 sampled permutations that violate their respective tolerance.

Discussion

- **Exact symmetry methods.** Canonisation, full symmetrisation, and linear S_N-equivariant layers are provably invariant; failures are zero within numerical precision.

Network	% non-invariant ↓	Pass?
Canonisation	0.00	✓
Full symmetrisation	0.00	✓
Sampled symmetrisation ($K = 256$)	0.94	✗
S_N-equivariant linear layers	0.00	✓
Learned-invariant (augmentations)	0.28	✓*

Table 1: Failure rate for each network.

- **Monte-Carlo method (c).** Sampled symmetrisation averages over $K = 0.05 \cdot 8!$ permutations— $\approx 0.05\,|S_N|$ for $N = 8$ - so estimator variance might explain the $\approx 1\%$ failure rate. Increasing K should decrease the error.
- **Learned invariance (e).** The unconstrained MLP acquires symmetry through augmentation; after ~ 250 epochs with ~ 500 augmentations per epoch the failure rate falls below 0.3%.

* Both stochastic models (c and e) pass with a looser tolerance, We intentionally kept a looser threshold here to reveal the *trend* toward invariance and to assess given our time and computation resources how far each model has progressed toward full invariance.